

AI Reverse Vending Machine - Project Report

Abstract

This project presents an AI-powered Reverse Vending Machine that detects recyclable objects using a laptop camera, awards eco-points, and generates coupon-based rewards with a PDF receipt.

Objectives

Develop a real-time recycling assistant using computer vision; encourage recycling through rewards; provide a modern user interface; create a scalable prototype.

Technologies

Python, Ultralytics RT-DETR, OpenCV, PyQt5, ReportLab, Flask (optional), Requests.

System Workflow

Camera captures an image, RT-DETR detects recyclable objects, the user confirms detection, points are calculated, a coupon is generated, and a PDF receipt is created.

AI Model

RT-DETR (Real-Time Detection Transformer) is a transformer-based object detection model that predicts object classes and bounding boxes directly without Non-Maximum Suppression, providing accurate real-time detection.

Modules

1. Camera Module
2. AI Detection Module
3. Reward System
4. Coupon Generator
5. PDF Receipt Generator
6. User Interface

Advantages

Fast detection, modern GUI, incentive-based recycling, modular architecture, and easy future integration with a mobile application.

Future Scope

Mobile app integration, cloud synchronization, user accounts, analytics dashboard, QR-code coupons, and smart city deployment.

Conclusion

The project demonstrates how AI and computer vision can promote sustainable recycling through an interactive reward-based system.